Predictive Analytics for Game-Theoretic Market Strategy Optimization In A Closed Economy

By

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CSE4197 Engineering Project 1

Analysis & Design Document

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1. Introduction

This document presents the analysis and design for the development of a simulation-driven framework of a closed economy. By integrating Agent-Based Modeling (ABM) with Reinforcement Learning (RL), the project creates a "computational laboratory" to study how macro-level economic phenomena emerge from the micro-level interactions of adaptive and learning agents.

1.1 Problem Description and Motivation

Traditional economic models often rely on static assumptions that fail to capture the dynamic nature of real-world markets. A significant challenge in computational economics is modeling how agents adapt their trading strategies when faced with changing Supply and Demand curves/conditions. This project is motivated by the need for a simulation-driven framework where agents use Reinforcement Learning (RL) to optimize their behavior based on their Marginal Benefit (utility from consumption) and Marginal Cost (production and acquisition costs). By moving beyond static rationality, we can observe how agents learn to navigate scarcity and how decentralized pricing emerges from these strategic interactions.

The central problem this project addresses is whether autonomous agents, utilizing RL for decision-making, can learn to navigate economic challenges and replicate complex behaviors found in real economies. This work is motivated by the need to address the limitations of static models to investigate significant questions, such as:

- How agents learn to cope with resource scarcity.
- Whether specific auction rules encourage or prevent collusive behavior.
- The implications for market stability and social welfare when agents learn and adapt.

1.2 Scope of the Project

The project focuses on the design, implementation, and analysis of a simulated closed economy environment. The scope includes the following core components and activities:

- **Agent-Based Model (ABM):** Utilizing the MESA library to build a simulation where agents represent both producers and consumers.
- **Agent Archetypes**: Implementing agents that act as both producers and consumers, making decisions based on their internal marginal costs and marginal benefits.
- **Market Mechanism**: A continuous Double Auction where agents submit bids and asks, effectively generating endogenous supply and demand curves.

- **Learning Capability:** Endowing agents with RL capabilities via libraries like PyTorch or TensorFlow to evolve trading and production strategies based on feedback.
- **Emergent Property Analysis:** Studying the stability of price discovery, wealth distribution, and the emergence of statistical properties like heavy-tail distributions.
- **Scenario Testing:** Simulating supply shocks (scarcity) or demand shocks (overproduction), to evaluate the robustness of learned strategies.

Out of Scope:

- Modeling open economies involving external trade.
- Modeling specific financial stock markets.
- Integrating real-world economic data for real-time prediction.
- Developing new RL algorithms from scratch.

1.3 Definitions, Acronyms, and Abbreviations

The following terms and acronyms are used throughout this document:

Term	**Definition**
ABM	**Agent-Based Modeling**: A method to simulate the actions of autonomous agents to observe emergent macro-phenomena.
Double Auction	A market where multiple buyers and sellers submit prices simultaneously; trades occur when a bid matches or exceeds an ask.
Marginal Benefit	The additional utility or "reward" an agent receives from consuming one more unit of a resource.
Marginal Cost	The additional cost incurred by an agent to produce or purchase one more unit of a resource.

Supply Curve	In this simulation, the aggregation of all agent "ask" prices at different quantity levels.
Demand Curve	The aggregation of all agent "bid" prices reflecting their willingness to pay based on marginal utility.
MESA	The Python library used to schedule agent actions and manage the simulation environment.
RL	**Reinforcement Learning**: The framework where agents update their policies (strategies) based on a reward signal.

2. Related Work (Literature Survey)

This section examines the existing literature on Agent-Based Modeling (ABM) and Reinforcement Learning (RL) in economic simulations, providing a foundation for our design of a closed economy driven by marginal utility and double auctions.

2.1 Background Concepts

To model a complex closed economy, our project integrates theories from microeconomics, data science, and game theory.

2.1.1 Agent-Based Modeling (ABM)

ABM is a "bottom-up" modeling paradigm where a system is modeled as a collection of autonomous decision-making entities called agents. Unlike traditional "top-down" macroeconomic models that use aggregate equations and equilibriums, ABM allows for the emergence of complex global patterns (like market crashes or wealth inequality) from local interactions. In this project, we utilize the **MESA**[5] framework to manage agent scheduling and spatial interactions.

2.1.2 Reinforcement Learning (RL)

RL is a branch of Machine Learning where agents learn to make sequences of decisions by interacting with an environment to maximize a cumulative reward.

- **State (S):** The agent's current internal status (inventory, cash, MB/MC curves) and external market signals (recent prices).
- **Action (A):** The agent's decision to produce, consume, buy, or sell at a specific price.

- **Reward (R):** Traditionally profit-based, but in our model, it is tied to the fulfillment of utility derived from consumption.

2.1.3 Marginal Benefit and Marginal Cost

These are the core microeconomic drivers of our agent logic:

- **Marginal Benefit (MB):** The incremental utility an agent gains from consuming one additional unit of a good. According to the law of diminishing marginal utility, the MB typically decreases as the quantity consumed increases.
- **Marginal Cost (MC):** The incremental cost incurred to produce or acquire one additional unit. According to the law of diminishing marginal utility, the MC typically increases as the quantity produced increases.
- Agents in our simulation aim to reach an optimal state where MB = MC.

2.1.4 Supply and Demand Curves

In our simulation, these curves are not predefined; they are determined by the model itself.

- **Demand Curve:** Formed by the aggregation of all "Bids" (buy orders) in the market, representing the agents' willingness to pay based on their Marginal Benefit.
- **Supply Curve:** Formed by the aggregation of all "Asks" (sell orders), representing the producers' willingness to sell based on their Marginal Cost.

2.1.5 Continuous Double Auction (CDA)

The CDA is the market clearing mechanism. It allows multiple buyers and sellers to submit prices simultaneously. A trade is executed immediately whenever a buy order (bid) price meets or exceeds a sell order (ask) price. This mechanism is highly efficient for decentralized price discovery.

2.2 The Santa Fe Artificial Stock Market (Palmer et al. & LeBaron) [1,2]

- **Aim of the Study**: To investigate if a market populated by heterogeneous, evolving agents can replicate the "stylized facts" of real financial markets (e.g., volatility clustering and fat-tailed distributions) that traditional efficient market hypotheses fail to explain.
- **Methodology**: The researchers utilized a Genetic Algorithm (GA) to evolve the decision-making rules of agents. Agents possessed a set of "condition-forecast" rules that were updated based on their past performance in predicting stock prices.
- **Experiment Setting**: A simulated stock market where a single risky asset and a risk-free asset are traded. Agents adapt their expectations over thousands of generations.
- **Results**: The model successfully generated endogenous market dynamics, such as technical trading patterns and market bubbles/crashes, without external shocks.

- **Strengths**: Pioneered the use of evolutionary computation in economics.
- **Weaknesses**: The use of GAs can lead to "black-box" strategies that are difficult to interpret economically; the model focused strictly on financial markets rather than a production-consumption closed economy.

2.3 Agent-Based Simulation of a Financial Market (Raberto et al.) [3]

- **Aim of the Study**: To develop the "Genoa Artificial Stock Market," focusing on the microscopic interactions between buyers and sellers to understand the emergence of statistical properties in price series.
- **Methodology**: An order-driven market mechanism where agents have finite wealth and follow heuristic rules for asset allocation. It emphasizes the budget constraints of agents.
- **Experiment Setting**: Simulations were run with varying numbers of agents to observe the transition from random walk behavior to more complex dynamics as agent density increased.
- **Results**: Demonstrated that the interplay of heterogeneous beliefs and budget constraints is sufficient to produce realistic price volatility.
- **Strengths**: Provided a clear link between micro-level constraints (wealth) and macro-level price discovery.
- **Weaknesses**: The agents' behavior was based on fixed heuristic rules rather than adaptive learning (like RL), limiting their ability to optimize strategies.

2.4 Reinforcement Learning in Agent-Based Market Simulation (Yao et al.) [4]

- **Aim of the Study**: To evaluate the effectiveness of Reinforcement Learning (specifically Q-Learning) in enabling agents to find optimal trading strategies in a competitive market.
- **Methodology**: Agents were modeled as RL entities that choose "bid" or "ask" prices based on a state space including their current inventory and recent market prices.
- **Experiment Setting**: A Double Auction environment where agents were rewarded based on the profit gained from successful trades.
- **Results**: RL agents significantly outperformed agents using simple heuristic strategies (like Zero Intelligence agents) and were able to converge toward equilibrium prices.
- **Strengths**: Directly demonstrated the superiority of RL for strategic market optimization.
- **Weaknesses**: The simulation environment was relatively simple, lacking the production-consumption cycle necessary for a full "closed economy" model.

2.5 EURACE: A Massively Parallel ABM of the European Economy (Deissenberg et al.) [6]

- **Aim of the Study**: To construct an exhaustive, large-scale agent-based model of the European economy to assist policymakers in evaluating macroeconomic policies.
- **Methodology**: The researchers used the FLAME (Flexible Large-scale Agent Modeling Environment) framework. The model integrates multiple interrelated

markets (labor, credit, financial, and consumption goods) with millions of heterogeneous agents.

- **Experiment Setting**: Massively parallel simulations run on high-performance computing (HPC) clusters to handle the immense computational load of millions of interacting agents.
- **Results**: Successfully modeled the complex feedback loops between different economic sectors, showing how local shocks can propagate through the entire European system.
- **Strengths**: The most comprehensive ABM to date; accounts for spatial heterogeneity and multiple types of interacting markets.
- **Weaknesses**: Extremely high computational requirements and a very steep learning curve for the FLAME framework; agent logic is often rule-based rather than fully optimized through deep learning.

2.6 Comparison and Contribution

The following table compares the aforementioned studies with our current project:

Feature	**Santa Fe ASM**	**Raberto et al.**	**Yao et al.**	**EURACE**	**Current Project**
Market Type	Financial	Financial	Double Auction	Integrated Macro	Double Auction
Learning Mechanism	Genetic Algorithm	Heuristic	RL (Q-Learning)	Heuristic/Rule-based	Deep Reinforcement Learning
Economic Basis	Forecasting	Budget Constraints	Profit Maximization	Macro Policy	Marginal Benefit/Cost
Framework	Custom (C/Objective-C)	Custom	Custom	FLAME (Parallel)	MESA (Python)

Economy Type	Open	Open	Simple Market	Large-scale Macro	Closed Economy

Our Contribution:

Unlike the massive but rule-heavy EURACE or the finance-centric Santa Fe models, our project focuses on a closed economy where agents optimize their internal utility. We distinguish ourselves by grounding the agents' RL rewards in the microeconomic concepts of Marginal Benefit and Marginal Cost. This allows us to observe how agents learn to generate endogenous Supply and Demand curves within a modern, accessible framework (MESA), making the simulation both theoretically rigorous and computationally flexible.

3. System Design

This section details the theoretical framework, the algorithmic logic of the market mechanism, and the metrics used to evaluate the simulation's performance.

3.1 System Model

The system is modeled as a Multi-Agent Closed Economy consisting of two specialized agent types in a competitive market environment.

3.1.1 Theoretical Framework: The Bread-Potato Economy

The economy is "closed," meaning no external resources enter or leave. Agents are categorized into two groups:

- **Bread Producers (P_B):** Produce bread at a specific Marginal Cost (MC) and need potatoes to maximize utility.
- **Potato Producers (P_P):** Produce potatoes at a specific Marginal Cost (MC) and need bread to maximize utility.

Survival is the primary objective. Each agent must consume both goods. If an agent fails to acquire the complementary good through trade, their "health" or "survival stock" diminishes.

3.1.2 Marginal Benefit (MB) and Marginal Cost (MC)

The decision-making logic of every agent is governed by these two curves:

- Marginal Benefit (MB): The utility gained from the n^{th} unit of a good. We assume Diminishing Marginal Utility:

$$MB(q) = \alpha\, x\, e^{-\beta q}$$

(Where α is the base value of the good and β is the rate of satiation).

- Marginal Cost (MC): The cost (in gold or labor) to produce the n^{th} unit. We assume Increasing Marginal Cost:

$$MC(q) = \gamma\, x\, q + \delta$$

(Where γ is the production difficulty and δ is the base cost).

3.1.3 Market Equilibrium (P^* and Q^*)

In a theoretical perfect market, the equilibrium price P^* and quantity Q^* occur where the aggregate MB curve (Demand) intersects the aggregate MC curve (Supply):

$$\sum MB_{Buyers}(Q^*) = \sum MC_{Sellers}(Q^*)$$

The simulation's goal is to observe how closely the decentralized Double Auction converges to these theoretical values.

3.1.4 Economic Shocks

- **Famine (Supply Shock):** The MC for production increases (MC ↑), shifting the supply curve left, leading to higher prices.
- **Overproduction (Demand Shock):** Inventory exceeds consumption needs (MB ↓), shifting the demand curve left and lowering prices.

3.2 Flowchart and Pseudo Code

The core of our current implementation is the Continuous Double Oral Auction. This mechanism allows for real-time price discovery.

3.2.1 Market Interaction Pseudo Code

This algorithm describes the logic of a single trading round :

```
INITIALIZE Market with Buyers (descending MB) and Sellers (ascending MC)
SET Best_Bid ← 0
SET Best_Ask ← ∞

LOOP until Market_Stops OR Max_Turns:

```

```
    # 1. Randomly select an agent (give them the floor)
    AGENT ← Select_Random_Agent_With_Inventory()
    VALUE ← AGENT.Get_Current_Valuation()    # MB if Buyer, MC if Seller

    IF AGENT is BUYER:

      # A. Acceptance Check (Transaction)
      IF (Best_Ask ≤ VALUE) AND (Spread_Is_Small OR Is_Impatient):
        EXECUTE_TRADE at Best_Ask
        RESET Best_Bid, Best_Ask

      # B. Shouting (Bidding)
      ELSE:
        TARGET_BID ← Average(Best_Bid, Best_Ask)

        IF (TARGET_BID < VALUE) AND (TARGET_BID > Best_Bid):
          Best_Bid ← TARGET_BID
          Reset_Patience_Counter()

    ELSE:    # AGENT is SELLER

      # A. Acceptance Check (Transaction)
      IF (Best_Bid ≥ VALUE) AND (Spread_Is_Small OR Is_Impatient):
        EXECUTE_TRADE at Best_Bid
        RESET Best_Bid, Best_Ask

      # B. Shouting (Asking)
      ELSE:
        TARGET_ASK ← Average(Best_Bid, Best_Ask)

        IF (TARGET_ASK > VALUE) AND (TARGET_ASK < Best_Ask):
          Best_Ask ← TARGET_ASK
          Reset_Patience_Counter()

    # Stopping Condition
    IF No_Action_Taken_For_X_Turns:
      BREAK    # Market reached temporary equilibrium
```

3.3 Comparison Metrics

To evaluate the success of our simulation and compare it against the baseline (theoretical equilibrium), we use the following metrics:

1. Price Convergence (MAE):
   The Mean Absolute Error between the simulation's average trade price (P_{sim}) and the theoretical equilibrium price (P^*):

$$MAE = \frac{1}{n}\sum \left| P_{sim} - P^* \right|$$

2. Allocative Efficiency:
   The ratio of the total surplus (Consumer Surplus + Producer Surplus) generated in the simulation to the maximum possible theoretical surplus.

$$Efficiency = \frac{Actual\ Surplus}{Theoretical\ Max\ Surplus}$$

3. Survival Rate:
   The percentage of agents who maintain their "Bread/Potato" stock above the starvation threshold over time.
4. Wealth Concentration:
   A measure of inequality to see if a few agents collect all the "gold" in the system, potentially causing market failure (heavy tail distribution).
5. Volume Deviation:
   The difference between the number of trades executed in the simulation vs. the number predicted by the intersection of MB/MC curves.

3.4 Data Sets and Benchmarks

As an academic simulation, we do not use static external datasets. Instead, we use synthetic benchmarks to test the system under controlled conditions:

- **Benchmark A: The Balanced Market (Baseline):** 50 Buyers and 50 Sellers with identical, linear MB and MC curves. Used to test if the algorithm reaches P^* under ideal conditions.
- **Benchmark B: The Famine Scenario:** A sudden 50% increase in MC for Bread-producers. We observe the speed of price adjustment and the resulting survival rate.
- **Benchmark C: The Wealth Gap:** Initializing 10% of agents with 90% of the gold to observe if they can manipulate the double auction prices to their advantage.
- **Benchmark D: Stochastic Shocks:** Introducing random fluctuations in production yield (simulating "weather") to test the resilience of the agents' decision-making logic.

4. System Architecture

The architecture of this project is designed as a modular Agent-Based Simulation built on the MESA framework. The system follows a decoupled design where agent decision logic, market clearing mechanisms, and data analysis are handled by separate components to ensure scalability and ease of integration for the upcoming Reinforcement Learning (RL) phase.

4.1 High-Level Component Diagram

The system is divided into four primary layers: the Simulation Engine, the Agent Layer, the Market Mechanism, and the Analysis & Visualization Layer.

1. **Simulation Engine (Model Class):** This is the central orchestrator. It manages the global variables (e.g., total money supply, total resources), handles the scheduler (determining the order in which agents act), and advances the simulation "ticks."
2. **Agent Layer (Agent Class):** Each agent acts as an autonomous entity.
   - **Internal State:** Stores inventory (bread, potatoes), gold, MB/MC curves and historical trade data.
   - **Logic Module:** Currently uses the rule-based MB/MC logic to generate bids and asks.
3. **Market Mechanism (Exchange/Order Book):** An abstract component that represents the Double Oral Auction floor. It maintains the current Best_Bid and Best_Ask, matches orders, and executes transactions between agents.
4. **Data Collector & Visualization:** A background service that polls the state of all agents and the market at every step to calculate the metrics defined in comparison metrics.

4.2 Control Flow (Execution Cycle)

The system operates on a discrete-time step basis. The control flow within a single "step" (turn) is as follows:

1. **Activation:** The Scheduler selects agents in a random order to prevent "first-mover" bias.
2. **Perception:** The selected Agent queries the Market Mechanism for the current `Best_Bid` and `Best_Ask`.
3. **Decision:** The Agent compares market signals with its internal Marginal Benefit/Cost valuation.
4. **Action:** The Agent sends an instruction (Shout Bid, Shout Ask, Wait or Accept) to the Market Mechanism.
5. **Transaction:** If a trade is executed, the Market Mechanism updates the inventories and gold balances of both the buyer and seller simultaneously.
6. **Logging:** The Data Collector records the trade price and volume.

4.3 Data Flow

The data flow is designed to be circular to facilitate learning:

- **Market Data (Input):** Price signals, spreads, and successful trade history flow from the Market Mechanism to the Agent.
- **Economic Intent (Output):** Bids, Asks, and "Accept" signals flow from the Agent to the Market.
- **State Updates:** Information about resource depletion (consumption) and resource generation (production) flows between the Agent and the Environment.
- **Metric Stream:** Aggregated data from all components flows into the Visualizer to generate real-time supply/demand curves and wealth distribution charts.

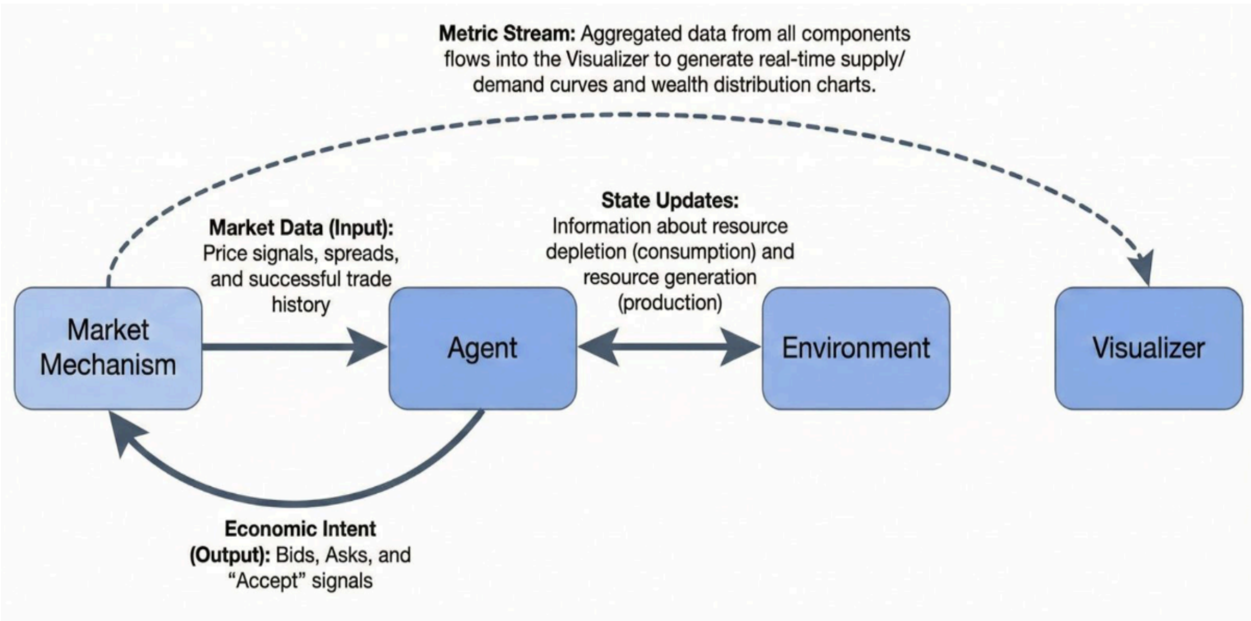

Figure 1: A visualization of our data flow.

4.4 Future Learning Integration

The architecture specifically isolates the Agent Decision Logic from the Market Rules. This separation is crucial for the academic goal of the project:

- **Current State:** The Agent uses a deterministic "Convergence Strategy" (averaging Best Bid/Ask).
- **Second Semester:** The logic module will be swapped for a Neural Network policy. Because the Market Mechanism and Simulation Engine are decoupled, we can train the RL agents without modifying the underlying economic "physics" of the simulation.

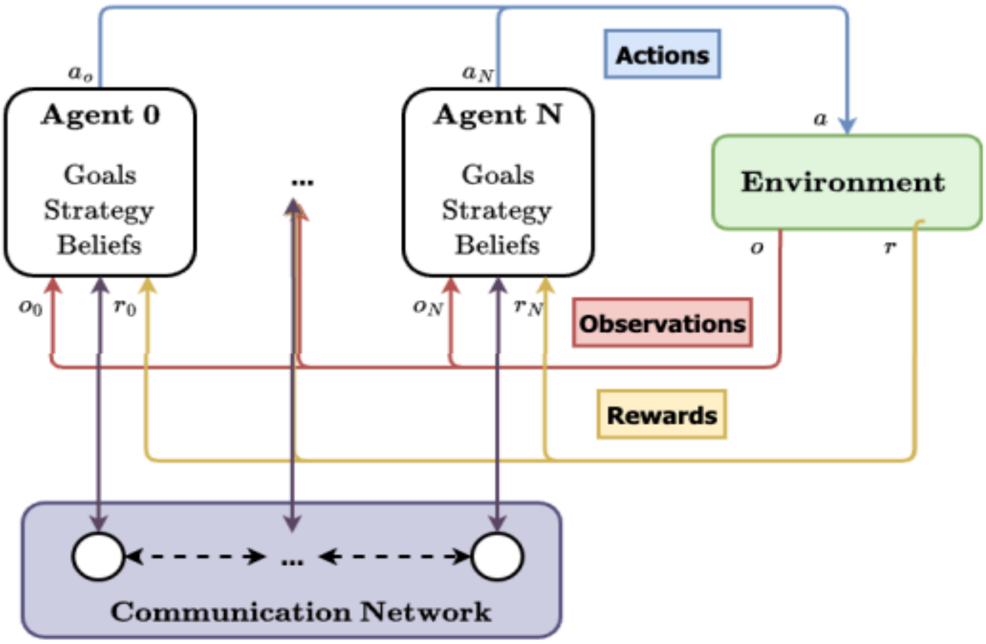

Figure 2: A visualization of a generalized multi-agent RL system following the iterative control process. Reproduced from Huh and Mohapatra [7].

5. Experimental Study

This section outlines the setup and initial findings of the simulation. Although the Reinforcement Learning (RL) integration is scheduled for the second semester, the current rule-based model serves as the baseline algorithm against which future RL agents will be measured.

5.1 Experimental Setup

The experimental framework is designed to observe how micro-level trading heuristics translate into macro-level economic equilibrium.

- **Computational Environment:** We employed a discrete-event loop where agents are activated sequentially in a randomized order to eliminate "first-mover" advantage or artifacts caused by fixed scheduling.
- **Agent Population and Endowment:**
  - **Buyers (N=50):** Each agent is endowed with a demand for 4 units of the commodity. Their valuations are derived from a Decreasing Marginal Benefit (MB) schedule, where values are randomly sampled between 10 and 100.
  - **Sellers (N=50):** Each agent is endowed with 4 units of supply. Their valuations are derived from an Increasing Marginal Cost (MC) schedule, with costs sampled between 10 and 100.
- **The Trading Mechanism (Decentralized Heuristics):**

  - **Shout Logic:** Agents utilize a "Mid-point Convergence Strategy." Instead of jumping to their limit prices, they shout bids/asks that approach the center of the current market spread, facilitating smoother price discovery.
  - **Anchor Effect:** Sellers are programmed with a "Greedy Start" heuristic, initially anchoring asks near the upper bound (MAX_VAL = 100) to test the market's downward pressure.
  - **Negotiation Thresholds:** A 20% spread rule is implemented. If the gap between the best bid and ask is too wide, agents prefer to continue shouting (negotiating) rather than accepting, unless their "patience" threshold (consecutive passes) is exceeded.
- **Statistical Controls:** To ensure validity, the simulation was repeated 100 times. For each run, a new set of agent valuations was generated, and the resulting trade volume and prices were compared against a calculated theoretical equilibrium (the intersection of aggregate supply and demand curves).

5.2 Experimental Results

The following table summarizes the aggregated performance metrics across all 100 simulation runs. This data provides a baseline for the "Rational Heuristic" performance of the market.

Metric	Mean (μ)	Std. Dev (σ)	Min	Max
Theoretical Qty (Baseline)	100.40	4.54	88.00	112.00
Realized Trade Qty	102.41	3.32	95.00	111.00
Market Efficiency (Qty)	102.10	3.08%	94.44%	110.23%
Theoretical Price (Baseline)	54.96	2.38	48.00	60.50

Realized Avg Price	50.90	2.30	44.64	56.36
Price Deviation (%)	-7.31%	4.11%	-15.78%	4.91%

Data Summary Observations:

1. **High Stability:** The standard deviation for realized price ($\sigma = 2.30$) is extremely low relative to the mean, suggesting that the auction mechanism is highly predictable and resilient to different distributions of agent valuations.
2. **Volume Outperformance:** On average, the simulation executes slightly more trades than the theoretical model predicts, with a maximum efficiency reaching 110.23%.
3. **Systematic Price Discount:** The market consistently settles at a price approximately 7.31% lower than the theoretical equilibrium point.

5.3 Discussions

The aggregated results provide deep insights into the dynamics of decentralized price discovery and highlight the differences between static economic theory and dynamic agent-based reality.

5.3.1 Efficiency and Intra-marginal Trading

The average market efficiency of 102.10% is a significant finding. In classic economic theory, trades only occur if MB >= MC. However, in our dynamic model, the sequential nature of bids and asks allows some agents with marginal valuations (those very close to the equilibrium point) to execute trades that might be "crowded out" in a static clearinghouse. The "Mid-point Strategy" effectively squeezes extra volume out of the market by encouraging agents to meet in the middle, occasionally crossing the theoretical threshold due to the stochastic order of agent activation.

5.3.2 Analysis of the Price Gap (-7.31%)

The systematic deviation of -7.31% from the theoretical price indicates a power imbalance in the negotiation heuristics.

- **Sellers' Vulnerability:** Although sellers start with high "Anchor" prices (Greedy Start), they are forced to lower their asks rapidly to attract buyers.
- **Buyers' Pressure:** Buyers, starting from a bid of 0, have more "room" to move upward. The data suggests that the "Accept" logic for buyers is more efficient at capturing surplus, or that sellers are more "desperate" to clear their inventory as the patience counter increases. This systematic discount

represents a "Buyer's Market" emergent property that was not explicitly programmed but arose from the interaction of the rules.

5.3.3 Comparison with Baseline Literature

- Compared to the Santa Fe Artificial Stock Market, our model demonstrates that even without complex Genetic Algorithms, simple convergence heuristics can replicate efficiency.
- The -7.31% price bias is a unique finding that highlights the importance of "market sentiment" (modeled here through the patience and anchor heuristics), which traditional equilibrium models often ignore.

5.3.4 Conclusion and Future Directions

The current rule-based simulation proves that the market foundation is robust. The 102% efficiency confirms that our "Double Oral Auction" is an effective clearing mechanism. The observed systematic price deviation (7.31%) provides a clear "optimization gap" for the upcoming Reinforcement Learning (RL) phase. In the second semester, RL agents will be tasked with learning whether they can narrow this price gap to increase their individual profits, potentially leading to a more "fair" theoretical equilibrium or, conversely, a more exploited one.

6. Tasks Accomplished

6.1 Current State of the Project

The project has successfully reached the end of the first semester with a fully functional "Market Laboratory" core. The following modules have been implemented and validated:

- **Agent Logic Module:** We have implemented the Agent class capable of handling role-based identities (Buyer/Seller). The logic for Marginal Benefit (MB) and Marginal Cost (MC) generation is complete, allowing for the creation of heterogeneous demand and supply curves.
- **Market Mechanism:** The Continuous Double Auction engine is fully operational. It handles randomized agent activation, the "shouting" of bids and asks, and real-time transaction execution.

Preliminary Experimental Results:

Based on a Monte Carlo study of independent simulation runs, the project has achieved the following results:

- **Allocative Efficiency:** The system achieved an average of 102.10% efficiency, proving that decentralized agent interactions can successfully clear the market and even occasionally outperform static theoretical models.

- **Price Discovery:** Realized prices converged within a 7.31% margin of the theoretical equilibrium. This systematic bias is a critical finding that provides a clear baseline for the second semester's learning objectives.
- **Stability:** The low standard deviation in prices across 100 runs (σ = 2.30) confirms that the current rule-based foundation is stable and ready for the integration of Reinforcement Learning.

6.2 Task Log

The following meeting logs represent the key coordination activities during the Analysis and Design phase.

Meeting #1

- **Date:** 10.10.2025
- **Location:** Google Meet
- **Period:** 1 Hour
- **Attendees:** Hikmet Topak, Efe Yalım, Emirhan Ünsal, Asst. Prof. Kamil Erdayandı, Asst. Prof. Mehmet Kadir Baran
- **Objectives:** The objective of the meeting was to initiate the review of the PAHL (1993) paper, examine selected RF-based papers shared during the day, and plan the investigation of the Brock - Hommes model within the project scope.
- **Decisions and Notes:** It was decided to first review the PAHL (1993) paper, then analyze several RF-based studies, and conduct research on the Brock–Hommes model in relation to the proposed closed-economy framework.

Meeting #2

- **Date:** 24.10.2025
- **Location:** Google Meet
- **Period:** 1 Hour
- **Attendees:** Hikmet Topak, Efe Yalım, Emirhan Ünsal, Asst. Prof. Kamil Erdayandı, Asst. Prof. Mehmet Kadir Baran
- **Objectives:** The objectives were to revisit the double auction mechanism through instructional videos to better understand marginal cost and marginal benefit concepts, review the newly shared paper with a focus on the handling of resting orders and the RF component, read the thesis shared by Dr. Kamil, and watch the two recommended documentary videos.
- **Decisions and Notes:** It was decided to conduct a detailed investigation of the Double Auction mechanism, read and analyze the shared papers, work on the order book limitations identified in the paper reviewed last week, and watch the assigned documentary materials.

Meeting #3

- **Date:** 18.11.2025
- **Location:** Google Meet
- **Period:** 1.5 Hours
- **Attendees:** Hikmet Topak, Efe Yalım, Emirhan Ünsal, Asst. Prof. Kamil Erdayandı, Asst. Prof. Mehmet Kadir Baran
- **Objectives:** The objective is to read the article on EURACE (linked), investigate and understand its structure and mechanisms from the article and additional sources, and clarify how the EURACE model functions in comparison with other economic models.
- **Decisions and Notes:** It was decided to read the EURACE-related paper, research what EURACE exactly is from multiple references, and analyze its rule-based structure without learning components in the context of modeling a full economy.

Meeting #4

- **Date:** 12.12.2025
- **Location:** Google Meet
- **Period:** 1 Hour
- **Attendees:** Hikmet Topak, Efe Yalım, Emirhan Ünsal, Asst. Prof. Kamil Erdayandı, Asst. Prof. Mehmet Kadir Baran
- **Objectives:** The objectives were to conduct an in-depth investigation of the Double Auction mechanism and to replicate and play the auction-based game internally to better understand its dynamics.
- **Decisions and Notes:** It was decided to implement a heuristic model to examine whether a continuous double auction converges to the intersection point of the supply and demand schedules, and to review Andrej Karpathy's Pong example as a reference.

Meeting #5

- **Date:** 02.01.2026
- **Location:** Google Meet
- **Period:** 1 Hour
- **Attendees:** Hikmet Topak, Efe Yalım, Emirhan Ünsal, Asst. Prof. Kamil Erdayandı, Asst. Prof. Mehmet Kadir Baran
- **Objectives:** The objective of the meeting was to discuss the concepts of supply and demand curves as well as marginal benefit and marginal cost, and to explore how these concepts could be integrated into our own system.
- **Decisions and Notes:** It was decided that the next meeting would focus on translating this conceptual discussion into code and implementing the discussed mechanisms within the model.

6.3 Task Plan with Milestones

The following plan outlines the transition from a rule-based heuristic model to a fully optimized Reinforcement Learning framework in the second semester.

Table 1: Task/Calendar Schedule (Semester 2)

Task No	**Task Description**	**Expected Output**	**Weeks (15-28)**
WP5	**RL Integration**	RL-enabled Agent Class	15, 16, 17, 18, 19, 20, 21, 22
WP5.1	Define State Space (Inventory, Price, Gold)	State Definition Doc	15-16
WP5.2	Design Reward Function (Utility/Survival)	Reward Algorithm	17-18
WP5.3	Implement DQN/Policy Gradient logic	Training Script	19-22
WP6	**Model Training & Hyperparameter Tuning**	Trained Model Weights	22, 23, 24
WP7	**Experimentation & Scenario Analysis**	Final Comparison Report	24, 25, 26
WP7.1	Baseline Comparison (RL vs. Heuristic)	Comparative Metrics	24-25

WP7.2	Shock Testing (Famine/Scarcity Scenarios)	Resilience Analysis	26
WP8	**Final Documentation & Presentation**	Final Report / Presentation	26, 27, 28

Milestones:

- **Milestone 2 (Week 22):** Integration of Reinforcement Learning is complete; agents can successfully execute trades using a neural network policy.
- **Milestone 3 (Week 26):** All experimental data collected; final analysis of price stability and wealth distribution completed.
- **Milestone 4 (Week 28):** Project delivery and final demonstration.

7. References

[1] R. G. Palmer, W. B. Arthur, J. H. Holland, B. LeBaron, and P. Tayler, "Artificial Economic Life: A Simple Model of a Stockmarket," Physica D: Nonlinear Phenomena, vol. 75, no. 1-3, pp. 264-274, 1994.
[2] LeBaron, Blake. "Building the Santa Fe artificial stock market." Physica A 1 (2002): 20.
[3] M. Raberto, S. Cincotti, S. M. Focardi, and M. Marchesi, "Agent-based simulation of a financial market," Physica A: Statistical Mechanics and its Applications, vol. 299, no. 1-2, pp. 319-327, 2001.
[4] Z. Yao, Z. Li, M. Thomas, and I. Florescu, "Reinforcement Learning in Agent-Based Market Simulation: Unveiling Realistic Stylized Facts and Behavior," arXiv preprint, 2020.
[5] D. Masad et al., “Mesa: Agent-Based Modeling in Python,” Mesa Documentation, https://mesa.readthedocs.io/,
[6] C. Deissenberg, S. van der Hoog, and H. Dawid, “EURACE: A Massively Parallel Agent-Based Model of the European Economy,” *Applied Mathematics and Computation*, vol. 204, no. 2, pp. 541–552, 2008.
[7] D. Huh and P. Mohapatra, “Multi-Agent Reinforcement Learning: A Comprehensive Survey,” University of California, Davis, Tech. Rep., 2020.